Usman Ali Khan

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Professional Summary

Fresh graduate and software engineer with strong analytical and problem-solving skills experienced in C# and .NET Core, focused on building and integrating reliable business systems, and eager to learn in collaborative teams.

Education

Asia Pacific University of Technology and Innovation — Kuala Lumpur, Malaysia
Bachelor of Science (Honours) in Computer Science — *July 2026*

- **Relevant Coursework:** Data Structures, Operating Systems, SDLC, Discrete Mathematics, Databases

Experience

Backend Engineer Intern — Jan. 2025 – May 2025
Bitlab Digital — *Kuala Lumpur, Malaysia*

- **Built and maintained backend APIs** using .NET Core (C#) integrated with Umbraco CMS, handling client-facing requests and front-end system integration.
- Implemented JWT-based authentication and authorization to **secure API endpoints against unauthenticated access** and payload manipulation.
- Built a **fault-tolerant dual-SMTP** email system with failover, monitoring performance and troubleshooting delivery issues.
- Designed and **maintained an SQL Server database** using Entity Framework Core, including ERD documentation, soft deletes, and iterative schema updates.
- Extracted and structured web content to **train an AI knowledge base** for a university companion project.

Projects

Aspectly (Video Generation SaaS) | *FastAPI, React, Remotion, Google Cloud Run* — Dec. 2025 – Present

- Building a video-creative **SaaS platform using FastAPI and React**, translating business requirements into features so users create, manage, and render branded video content.
- Developing a **cloud-based video render** pipeline using Remotion and Google Cloud Run that automatically processes and delivers rendered video assets at scale, deploying dedicated worker services monitoring performance and debugging failures.

CubeSat Control (Real-Time System Simulation) | *Rust, Tokio, TCP* — Jul. 2025 – Sep. 2025

- Developed a high-performance **real-time satellite control simulation in Rust**, featuring an onboard control system (OCS) and a ground control station (GCS) for telemetry and command exchange.
- Implemented **lock-free data structures** and priority-based task scheduling with CPU affinity, testing and **monitoring performance** for deterministic, sub-millisecond jitter.

Fruit Factory Production Simulator (Course Project) | *Java* — Aug. 2024 – Nov. 2024

- Built a **multi-threaded Java simulation** of a factory pipeline using OOP principles, with each stage modeled as its own class running on a separate thread and communicating through shared, bounded buffers.
- Used locks and monitors to **coordinate threads safely**, prevent race conditions, and avoid deadlocks when resources were limited.
- **Simulated real-world failures** (equipment breakdowns, delays) to test and monitor performance under contention and recovery.

Technical Skills

Languages: C#, C++, Java, SQL,Python, Javascript, HTML, CSS
Frameworks / Tools: GitHub, FastAPI, Linux, .NET Core, React, Docker, Databases, GCP, Firebase, Agentic Development Tools
Speaks: English (IELTS 7.5), Urdu / Hindi (Native)

Extra-Curricular Activities

- Led a team of 2 as **Event Coordinator Lead** for APU Red Hat Club, planning and executing 5+ events (40–150 participants) while managing logistics, timelines, and cross-team coordination to improve outreach.
- Served as **Peer Leader for Mathematics for Computing**, tutoring students and supporting their understanding of core concepts.